

Summary

Results-driven software engineer with 6+ years of experience developing secure, reliable, and cloud-native applications in payments, compliance, and healthcare domains. Proven ability to design and deliver highly available distributed systems, ensure compliance with GDPR/CCPA, and implement CI/CD automation for faster delivery cycles. Skilled in Java, Spring Boot, REST, microservices, SQL/NoSQL, AWS, and containerized deployments. Adept at mentoring engineers, mitigating risks for high-visibility applications, and driving operational excellence with a never-down strategy.

Education

Master of Science in Computer Software Engineering

GPA 3.72

May 2019

Arizona State University, Tempe, AZ

Courses: Introduction to Programming Languages, Introduction to Software Engineering, Advanced Data Structures and Algorithms, Mobile Computing, Software Design, Software Verification/Validation/Testing, Software Project/Process and Quality Management, Software Factory, Data Mining

Bachelor of Engineering in Computer Science

GPA 3.51

May 2017

The Maharaja Sayajirao University of Baroda, Vadodara, India

Courses: OOPS with C++, Java, Advanced DBMS, Operating Systems, Network Security, Client Server Architecture, ASP.NET

Technical Skills

- **Languages:** C, C++, JAVA, C#.NET, SQL, HTML, CSS, Python, JavaScript, Node.js
- **Frameworks:** Spring Boot, Hibernate, React.js, Express.js, AngularJS, Flask, Bootstrap, Spring Cloud, Reactor, WebFlux, Spring Security, JPA
- **Tools:** Git, Maven, Jira
- **Databases:** Oracle, MySQL, MS SQL SERVER, PostgreSQL, MongoDB, Couchbase, DynamoDB, Redis
- **Cloud & DevOps:** AWS(Lambda, EC2, ECS, S3, CloudFormation, IAM, RDS), Docker, Jenkins, Kubernetes, CodePipeline
- **Messaging:** Kafka, SNS, SQS, Webhooks, RabbitMQ
- **Monitoring:** CloudWatch, Prometheus, Grafana, OpenTelemetry, ELK Stack, Zipkin, Actuator
- **AI:** OpenAI, Prompt Engg., GitHub Copilot, GPT-4, Dall.E, LLM, LLMops, Text2SQL, FastMCP, ADK
- **Practices:** Microservices, DDD, Event-driven architecture, CI/CD, 12-Factor App, Security & Compliance (GDPR/CCPA/PCI), Performance Tuning, REST API Design, Test Automation

Experience

Software Development Engineer

November 2020 – Present

Amazon.com Services LLC, Seattle, Washington

- **Architected and delivered** large-scale **Java/Spring Boot** microservices using **RESTful APIs, Spring Cloud, and AWS** (EC2, ECS, Lambda, API Gateway, SQS, SNS, DynamoDB, RDS, CloudFormation) to automate complex accounting and compliance workflows across multiple business units.
- **Migrated monolithic services** to a **containerized microservices architecture** using **Docker and AWS ECS**, improving release frequency by **60%** and simplifying scalability for peak traffic events.
- **Implemented event-driven design patterns** with **Kafka, SQS, and SNS**, enabling asynchronous communication, decoupled service interactions, and improved system resilience under high concurrency.
- **Optimized database performance** across **PostgreSQL, DynamoDB, and Redis** by tuning queries, partitioning tables, and leveraging caching strategies to reduce latency by **35%** for high-traffic APIs.
- Designed and maintained **API contracts and schema validation** (OpenAPI/Swagger, JSON Schema) to ensure backward compatibility and enforce data quality across microservices.
- **Integrated observability** using **AWS CloudWatch, Prometheus, Grafana, and OpenTelemetry**, defining SLIs/SLOs and alerting pipelines that reduced MTTR by **40%**.
- **Enhanced CI/CD pipelines** with **Jenkins, Maven, GitHub Actions, and AWS CodePipeline**, achieving **zero-downtime deployments** and improving rollback safety using blue-green strategies.
- Collaborated with **cross-functional teams** (DevOps, Security, Data Engineering) to define architecture standards, adopt **12-Factor App** principles, and ensure compliance with **GDPR, CCPA, PCI DSS**.
- Implemented **distributed caching (ElastiCache/Redis)** and **rate-limiting middleware**, improving API throughput under high load and enhancing user experience for global customers.
- Conducted performance profiling and load testing using **JMeter, Gatling, and AWS X-Ray**, identifying critical bottlenecks and enabling proactive scaling policies.
- Drove **code quality and reliability** initiatives — integrated **SonarQube**, improved test coverage to 85%, and introduced **contract testing with Pact** for safer inter-service deployments.

- **Led technical design reviews and mentored engineers** across teams, guiding best practices in asynchronous programming, reactive patterns (Project Reactor), and domain-driven design (DDD).
- Partnered with the **SRE organization** to achieve 99.99% service availability and align operational metrics with business SLAs through runbooks, chaos testing, and canary rollouts.
- Contributed to the **AWS Well-Architected Framework** adoption, improving security posture and cost efficiency via tagging, auto-scaling groups, and IAM least-privilege enforcement.
- Championed innovation by **evaluating emerging tools** (Quarkus, Micronaut, gRPC, GraphQL) and conducting POCs to modernize the microservices stack.

Java Application Engineer

September 2019 - November 2020

HCA Healthcare, Nashville, Tennessee

- **Designed and implemented RESTful APIs and backend services** using **Java 11, Spring Boot, and Spring MVC** to support 1,500+ staff across 500+ healthcare facilities, enabling secure registration, patient alerts, and real-time operational tracking.
- **Engineered reactive microservices** using **Spring WebFlux and Project Reactor**, achieving non-blocking request handling and improving system responsiveness by **45%** under concurrent load.
- Integrated services with **Couchbase, SQL Server, and PostgreSQL**, designing efficient **data access layers (JPA/Hibernate)** and optimizing queries for high availability and secure persistence of patient metrics and alerts.
- **Developed an event-driven notification framework** that streamed alerts to **email, Webex Teams, and Slack channels** via **Webhooks and Kafka**, reducing incident resolution times for care staff.
- **Implemented authentication and authorization** using **Spring Security and OAuth 2.0**, enforcing RBAC policies and HIPAA-compliant data access controls.
- Created **custom monitoring and alerting pipelines** leveraging **Prometheus, Grafana, and ELK stack**, providing real-time visibility into API performance and system health.
- **Automated deployments** through **Jenkins pipelines, Dockerized containers, and AWS EC2 instances**, reducing manual release effort by **80%** and standardizing deployment environments.
- Collaborated with frontend teams using **React.js** and **AngularJS** to design and document REST contracts via **OpenAPI/Swagger**, improving API adoption and reducing integration bugs by **30%**.
- **Enhanced application observability** using **Spring Actuator** and integrated distributed tracing via **Zipkin**, improving root-cause analysis speed by **50%**.
- **Implemented caching strategies** (Redis and Spring Cache) and **asynchronous message handling** to reduce database load and improve API response time during peak usage.
- **Participated in Agile ceremonies and sprint planning**, leading backlog grooming and code review sessions to ensure code quality, maintainability, and design consistency.
- **Enforced secure data handling practices** ensuring HIPAA compliance and patient privacy using **encryption at rest (AES-256) and TLS in transit**.
- **Developed unit and integration tests** using **JUnit, Mockito, and Postman collections**, raising test coverage to **85%+** and ensuring stable continuous delivery.
- **Partnered with DevOps and Security teams** to integrate static code analysis (SonarQube) and vulnerability scanning (OWASP Dependency Check) into CI/CD pipelines.
- **Modernized legacy workflows** into distributed, event-driven services using **Spring Cloud, Kafka, and RESTful APIs**, improving reliability and cutting payment method launch time by **8–12 weeks**.
- Delivered multiple enhancements for **patient workflow management systems**, directly contributing to reduced check-in time and improved patient satisfaction metrics.

Projects

FACE RECOGNITION APPLICATION

- The website allows a user to **detect** faces in an image. The website also keeps the track of the number of face detections a user has performed through his/her account on the website.
- The **frontend** of the website is developed using **HTML, CSS, JavaScript, React.js, JSX** and **Redux.js**.
- The **backend server** is developed using **Node.js** and **Express.js**. For storing the user data, **PostgreSQL** has been used as the database.
- The website has been deployed on **Heroku**: <https://face-recognition-vb.herokuapp.com/>.